

Evidential Dirichlet Fusion for Trusted Multimodal Spectrum Sensing

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Abstract—Reliable spectrum sensing under low signal-to-noise ratio (SNR) and unseen modulation conditions remains challenging because existing deep learning detectors often rely on a single representation and produce overconfident decisions under distribution shifts. This paper proposes an uncertainty-aware multimodal framework for robust binary spectrum sensing. It fuses complementary temporal and signal-derived image representations through gated cross-attention and employs Dirichlet evidential learning to jointly estimate spectrum-occupancy probabilities and epistemic uncertainty. Experiments on the RadioML 2018.01A dataset cover in-distribution validation, unseen-SNR testing, and unseen-modulation testing. Experimental results demonstrate that the proposed method consistently outperforms representative single-modal and multimodal baselines under all three evaluation protocols, achieving the highest accuracy, F1-score, and AUC in each case while maintaining a low false-alarm rate and robust performance under unseen-SNR and unseen-modulation conditions. These results demonstrate that uncertainty-aware multimodal evidence fusion provides a more reliable and trustworthy solution for practical spectrum sensing under distribution shifts.

Keywords—Spectrum sensing, cognitive radio, multimodal learning, evidential deep learning, uncertainty estimation.

I. INTRODUCTION

A. Background and Motivation

Spectrum scarcity remains a fundamental challenge in modern wireless systems. The rapid growth of cognitive radio, dynamic spectrum access, the Internet of Things, and next-generation wireless networks has increased the demand for limited radio resources, making static spectrum allocation increasingly inefficient. Spectrum sensing enables secondary users to identify temporarily vacant bands without causing harmful interference to licensed users and is therefore a key component of dynamic spectrum access [1]–[3].

Binary spectrum sensing is commonly formulated as a hypothesis-testing problem, where H_0 denotes an idle channel containing only noise and H_1 denotes an occupied channel containing both signal and noise. A practical detector should identify weak signals while maintaining a low false-alarm probability. However, propagation loss, fading, interference, hardware impairments, noise uncertainty, and modulation diversity can substantially distort received observations. Moreover, the SNR ranges and modulation formats encountered during deployment may differ from those used for training. Spectrum sensing should therefore be evaluated not only under in-distribution conditions, but also under signal-distribution shifts [3], [4].

B. Limitations of Prior Work

Conventional methods, including energy detection, matched filtering, and cyclostationary detection, rely on sta-

tistical assumptions or prior signal knowledge. Energy detection is sensitive to noise uncertainty, matched filtering requires accurate waveform information and synchronization, and cyclostationary detection often introduces considerable computational cost [3], [5].

Learning-based approaches reduce dependence on hand-crafted decision rules by extracting discriminative features from IQ sequences, spectral representations, or signal-derived images. CNN-based methods capture spectral and spatial structures, whereas recurrent networks model temporal dependencies [6]–[8]. Nevertheless, most existing detectors rely on a single representation. Temporal features may be severely corrupted by noise, while image-based representations may lose fine-grained waveform dynamics. Their complementary characteristics motivate joint temporal–image modeling.

BCAM-Net combines magnitude–phase temporal features with CWT-based time-frequency representations through bidirectional cross-attention [9]. Although it improves multimodal interaction under generalized Gaussian noise, the cross-modal updates are not explicitly regulated before being incorporated into the fused representation. Its Softmax classifier also produces deterministic probabilities without directly representing the evidence supporting each prediction.

Reliable confidence estimation is particularly important under distribution shifts. Neural networks can remain highly confident even when their predictions are incorrect [10], and this problem may become more severe for unseen signal conditions [4]. Post-hoc calibration depends on validation data, while Bayesian approximations such as Monte Carlo dropout require repeated inference [10], [11]. Evidential deep learning instead predicts class evidence and parameterizes a Dirichlet distribution, allowing low-evidence observations to produce higher uncertainty [12]. This property is well suited to weak, ambiguous, and previously unseen signals.

C. Challenges and Solutions

To address these limitations, this paper proposes an evidential multimodal framework for robust binary spectrum sensing. Complementary temporal and signal-derived image representations are first extracted from the same IQ observation. A gated bidirectional cross-modal refinement module then projects information between the two modalities and regulates the residual updates using learnable scalar gates. Finally, a Dirichlet evidential classifier jointly produces spectrum-occupancy probabilities and evidence-based uncertainty.

The main contributions are summarized as follows:

- **Dual-branch multimodal representation:** Temporal waveform dynamics and multiple signal-derived image structures are jointly modeled to provide complementary descriptions of weak and unfamiliar signals.

- **Gated bidirectional cross-modal refinement:** Bidirectional feature projections and learnable scalar gates are introduced to control cross-modal residual updates and reduce the influence of unreliable transferred information.

- **Evidential decision modeling:** Dirichlet evidence is used to jointly estimate spectrum-occupancy probabilities and epistemic uncertainty, reducing unwarranted confidence under unfamiliar signal conditions.

- **Evaluation under distribution shifts:** Experiments on RadioML 2018.01A evaluate in-distribution performance and generalization to unseen SNRs and modulation formats. The proposed method consistently improves ACC, F1-score, and AUC while maintaining a low false-alarm probability.

II. RELATED WORK

A. Conventional Spectrum Sensing

Conventional spectrum-sensing methods infer spectrum occupancy from statistical or structural properties of received signals. Energy detection requires neither waveform knowledge nor synchronization, but is highly sensitive to noise uncertainty, particularly at low SNRs [3]. Matched filtering offers high detection sensitivity with short observations, although it requires accurate waveform and synchronization information [3]. Cyclostationary detection distinguishes communication signals from stationary noise by exploiting periodic statistical structures; multiple cyclic frequencies can improve robustness at the cost of increased observation time and computational complexity [5]. Overall, these methods remain effective when their assumptions are satisfied, but their performance degrades in blind and distribution-varying environments.

B. Learning-Based Spectrum Sensing

Learning-based methods replace fixed analytical statistics with data-driven feature extraction and classification. Early approaches relied on handcrafted energy, covariance, entropy, or spectral features, whereas recent deep models learn representations directly from sensing observations.

Lee *et al.* developed a CNN-based cooperative sensing method to capture correlations among observations from multiple secondary users [6]. Tekbiyik *et al.* combined spectral correlation functions with CNNs for joint spectrum sensing and signal identification [7]. Kumar *et al.* employed an RNN-BiLSTM architecture to model temporal dependencies of NOMA waveforms under diverse channel conditions [8].

Image-based methods further transform received signals into structured representations for convolutional processing. STFT-CNN and wavelet-ResNet exploit localized and multiscale time-frequency patterns, respectively [13], [14], while Spectrum Transformer captures long-range dependencies in wideband observations [15]. Despite their effectiveness, most existing detectors rely on a single representation and may therefore lose complementary temporal, phase, or frequency-domain information. Their robustness to unseen SNRs and modulation formats also remains insufficiently studied.

C. Multimodal Spectrum Sensing

Multimodal spectrum sensing combines heterogeneous signal views to reduce the information loss associated with single-representation detectors. Gao *et al.* integrated sequential features and CWT-based spectrograms using a GAN-GRU-YOLO framework for consumer-IoT spectrum sensing [16]. Fang *et al.* combined CNN and Transformer components to model local observations and long-range cooperative-user dependencies [17].

BCAM-Net jointly models magnitude-phase temporal features and CWT-based time-frequency representations through bidirectional cross-attention [9]. Its mutual token-level interaction improves low-SNR sensing, but the transferred cross-modal information is incorporated without an additional mechanism for controlling the magnitude of the feature updates. Moreover, its deterministic Softmax classifier does not explicitly quantify the evidence supporting each decision.

Evidential deep learning maps nonnegative class evidence to a Dirichlet distribution, enabling class probabilities and evidence-based uncertainty to be estimated jointly [12]. However, its integration with controlled bidirectional multimodal refinement remains largely unexplored in spectrum sensing. These observations motivate the proposed framework, which combines complementary temporal and signal-derived image representations, gated bidirectional cross-modal refinement, and evidential decision modeling for reliable sensing under distribution shifts.

III. FRAMEWORK OVERVIEW

A. Algorithm Overview

The proposed framework is developed for reliable binary spectrum sensing under low signal-to-noise ratio (SNR) conditions and signal-distribution shifts. Given a complex baseband IQ observation $\mathbf{x} = [\mathbf{I}, \mathbf{Q}] \in \mathbb{R}^{N \times 2}$, where $N = 1024$, the sensing task is formulated as

$$\mathbf{x}[n] = \begin{cases} \mathbf{w}[n], & H_0, \\ \mathbf{s}[n] + \mathbf{w}[n], & H_1, \end{cases} n = 1, \dots, N, \quad (1)$$

where $\mathbf{s}[n]$ denotes the received primary-user signal and $\mathbf{w}[n]$ denotes additive noise. Under H_0 , the observed spectrum segment is regarded as idle, whereas under H_1 , a primary-user signal is present. The objective is to detect weak occupied-spectrum observations while maintaining a low false-alarm probability.

Before multimodal representation construction, each IQ observation is normalized by its average power:

$$\bar{\mathbf{x}} = \frac{\mathbf{x}}{\sqrt{\frac{1}{2N} \sum_{n=1}^N (I^2[n] + Q^2[n]) + \varepsilon}}, \quad (2)$$

where ε is a small positive constant introduced for numerical stability.

As illustrated in Fig. 1, the proposed framework comprises four main stages. First, the normalized IQ sequence is converted into six-channel temporal and eight-channel signal-derived image representations. Second, a temporal convolutional encoder and a residual image encoder extract complementary embeddings from the two modalities. Third, a gated bidirectional cross-modal refinement module transfers information between the embeddings while regulating residual-update strength. Finally, the original and refined representations are processed by a trainable evidential decision head to produce nonnegative class evidence, spectrum-occupancy probabilities, and an evidence-based uncertainty estimate.

Unlike conventional dual-branch architectures that directly concatenate independent features, the proposed framework performs controlled bidirectional feature refinement before decision making. Moreover, the evidential head distinguishes relative class preference from the amount of evidence supporting the prediction. All trainable components, including the two feature encoders, cross-modal refinement module, decision MLP, and evidential output layer, are jointly optimized through the end-to-end objective defined in Section III-F.

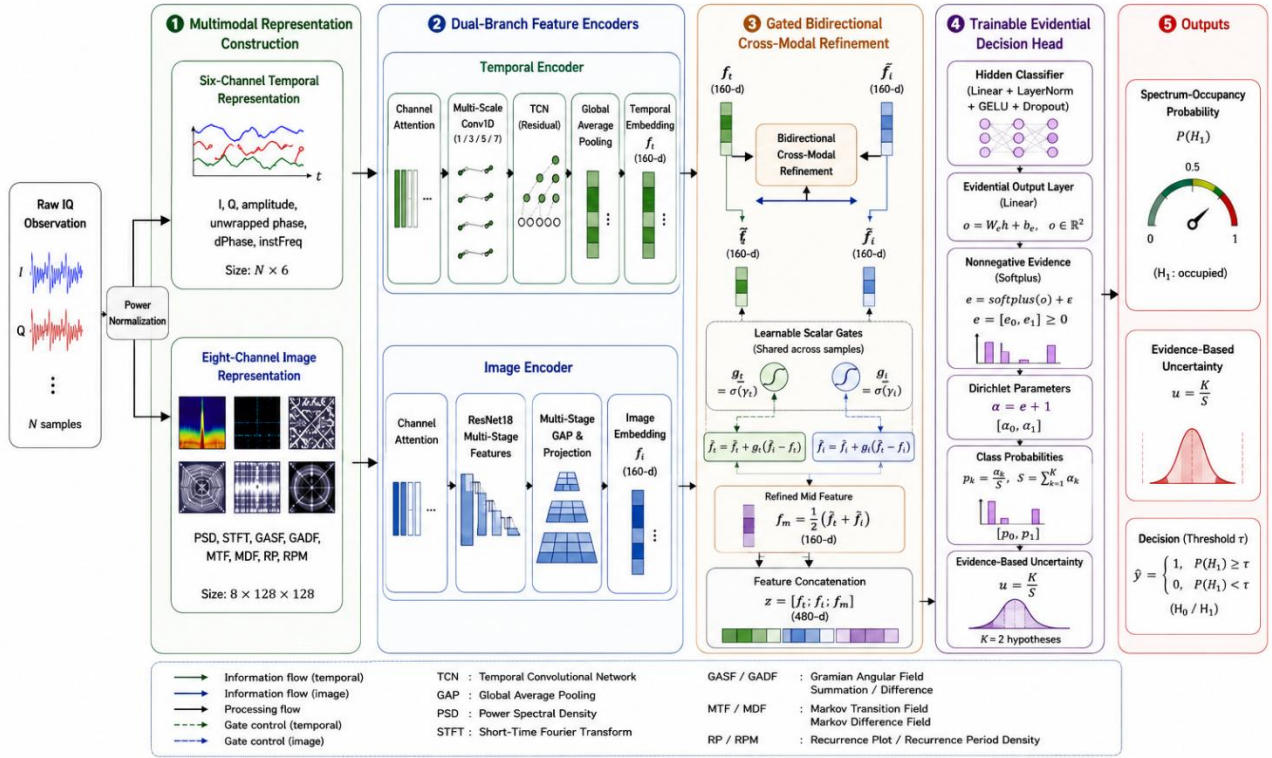


Fig. 1. Overall architecture of the proposed uncertainty-aware multimodal spectrum-sensing framework. A raw IQ observation is power-normalized and converted into complementary temporal and signal-derived image representations. The two representations are processed by dedicated feature encoders, refined through gated bidirectional cross-modal interaction, and mapped to Dirichlet evidence for spectrum-occupancy prediction and uncertainty estimation.

B. Multimodal Signal Representation

(1) Temporal Representation

The normalized IQ observation is converted into a six-channel temporal sequence consisting of the I and Q components, instantaneous amplitude, unwrapped phase, phase difference, and normalized instantaneous frequency. The phase difference and normalized instantaneous frequency are defined as

$$\begin{aligned}\Delta\phi[n] &= \phi[n] - \phi[n-1], \\ f_{\text{inst}}[n] &= \text{clip}\left(\frac{\Delta\phi[n]}{2\pi}, -1, 1\right).\end{aligned}\quad (3)$$

The resulting temporal input is expressed as

$$\mathbf{X}_t = [\mathbf{I}, \mathbf{Q}, \mathbf{A}, \boldsymbol{\phi}, \boldsymbol{\Delta\phi}, \mathbf{f}_{\text{inst}}] \in \mathbb{R}^{N \times 6}. \quad (4)$$

Each channel is independently standardized using the mean and standard deviation of the current observation.

(2) Signal-Derived Image Representation

The same normalized IQ observation is transformed into an eight-channel image tensor

$$\mathbf{X}_i = \text{Stack}(\mathbf{P}_{\text{PSD}}, \mathbf{P}_{\text{STFT}}, \mathbf{G}_S, \mathbf{G}_D, \mathbf{M}_T, \mathbf{M}_D, \mathbf{R}_P, \mathbf{R}_{PM}) \in \mathbb{R}^{8 \times H \times W}, \quad (5)$$

where $H = W = 128$. The eight channels are organized into four complementary groups describing spectral characteristics, pairwise temporal relations, amplitude-state transitions, and recurrent structures.

The spectral group consists of the PSD and STFT channels. Welch's method is applied to the complex sequence $I + jQ$ using a segment length of 128 and an overlap of 64. After logarithmic scaling and min-max normalization, a two-dimensional PSD relation map is constructed as

$$\mathbf{P}_{\text{PSD}} = \tilde{\mathbf{p}} \tilde{\mathbf{p}}^T, \quad (6)$$

where $\tilde{\mathbf{p}}$ denotes the normalized log-PSD vector. The STFT channel is computed from the same complex sequence using

a window length of 64 and an overlap of 32. Its log-magnitude spectrogram is min-max normalized and resized to 128×128 .

For the remaining channels, the instantaneous-amplitude sequence is uniformly downsampled from 1024 to 128 samples. The downsampled sequence is first rescaled to $[-1, 1]$, and the GASF and GADF representations are then constructed following the standard Gramian angular field formulation [19]. The two resulting maps are independently min-max normalized to $[0, 1]$.

The same downsampled amplitude sequence is further partitioned into eight discrete states using quantile-based discretization, from which the MTF representation is constructed [18]. To emphasize the relative transition distribution within each row of the MTF image, row-wise normalization is first performed as

$$\hat{M}_D(m, n) = \frac{M_T(m, n)}{\sum_j M_T(m, j) + \varepsilon}, \quad (7)$$

where ε is a small positive constant introduced for numerical stability. The resulting map is then min-max normalized:

$$\mathbf{M}_D = \text{MinMax}(\hat{\mathbf{M}}_D). \quad (8)$$

In this work, the resulting MTF-derived representation is denoted by MDF.

Finally, two recurrence maps are generated from the same downsampled amplitude sequence using recurrence rates of 20% and 5% [19]:

$$R_\rho(m, n) = \mathbb{I}(|A[m] - A[n]| \leq \varepsilon_\rho), \quad (9)$$

$$\rho \in \{5\%, 20\\},$$

where ε_ρ is selected according to the prescribed recurrence rate. The two settings form the RP and RPM channels, respectively, and characterize relatively dense and sparse recurrence

structures.

The six continuous-valued channels are min-max normalized to $[0,1]$, whereas RP and RPM remain binary. All eight maps are finally arranged as 128×128 images and stacked along the channel dimension to form the final $8 \times 128 \times 128$ image tensor.

C. Dual-Branch Feature Extraction

(1) Temporal Feature Encoder

The temporal encoder first applies a channel-attention module to adaptively recalibrate the six temporal channels. A parallel multi-scale one-dimensional convolutional front end with kernel sizes of 1, 3, 5, and 7 is then employed to extract waveform features under different local receptive fields. The outputs of the four branches are concatenated along the channel dimension to form a 64-channel feature map, which is subsequently processed by two residual Temporal Convolutional Network (TCN) blocks with dilation factors of 2 and 4 [20]. Finally, global average pooling over the temporal dimension produces the temporal embedding

$$\mathbf{f}_t = E_t(\mathbf{X}_t) \in \mathbb{R}^d, d = 160. \quad (10)$$

(2) Image Feature Encoder

The image encoder first applies a channel-attention module to adaptively recalibrate the eight signal-derived image channels. The reweighted image tensor is then processed by a ResNet18 backbone whose first convolutional layer is modified to accept eight-channel inputs [21].

To aggregate image features from different representation levels, the outputs of the second, third, and fourth residual stages are retained and denoted by $\mathbf{F}_2$, $\mathbf{F}_3$, and $\mathbf{F}_4$, respectively. After global average pooling, the resulting feature vectors have dimensions of 128, 256, and 512. They are concatenated along the feature dimension to form a multi-stage representation:

$$\mathbf{f}_{\text{pyr}} = [\text{GAP}(\mathbf{F}_2); \text{GAP}(\mathbf{F}_3); \text{GAP}(\mathbf{F}_4)] \in \mathbb{R}^{896}. \quad (11)$$

A linear projection layer then maps the concatenated representation into the shared embedding space used by the temporal branch:

$$\mathbf{f}_i = E_i(\mathbf{X}_i) = \text{Proj}(\mathbf{f}_{\text{pyr}}) \in \mathbb{R}^d, d = 160 \quad (12)$$

This multi-stage aggregation allows the image encoder to jointly exploit relatively fine-grained local structures and higher-level global representations.

D. Gated Bidirectional Cross-Modal Refinement

The temporal encoder and image encoder extract complementary features from the same IQ observation. Although direct concatenation preserves the original embeddings of both modalities, it does not allow information from one modality to refine the other before classification. Meanwhile, uncontrolled cross-modal information transfer may introduce noise or modality-specific interference into the fused representation. Therefore, a gated bidirectional cross-modal refinement module is introduced at the embedding level to explicitly regulate the information exchange between the two modalities. Prior multimodal studies have shown that appropriately constraining cross-modal information transfer can improve fusion stability [22], [23].

Since both branches complete global pooling before fusion, the cross-modal interaction is directly performed on the temporal embedding $\mathbf{f}_t$ and the image embedding $\mathbf{f}_i$. The context vectors in the two directions are computed as

$$\begin{aligned} \mathbf{c}_{t \leftarrow i} &= \mathbf{W}_{i \rightarrow t} \mathbf{f}_i, \\ \mathbf{c}_{i \leftarrow t} &= \mathbf{W}_{t \rightarrow i} \mathbf{f}_t. \end{aligned} \quad (13)$$

where $\mathbf{W}_{i \rightarrow t}, \mathbf{W}_{t \rightarrow i} \in \mathbb{R}^{d \times d}$, are learnable linear projection matrices. The context vector $\mathbf{c}_{t \leftarrow i}$ maps structural information from the image modality into the temporal embedding space, whereas $\mathbf{c}_{i \leftarrow t}$ maps temporal-evolution information into the image embedding space.

To control the overall strengths of the cross-modal updates in the two directions, two learnable scalar gates are introduced:

$$g_t = \sigma(\gamma_t), g_i = \sigma(\gamma_i), \quad (14)$$

where γ_t and γ_i are trainable parameters, and $\sigma(\cdot)$ denotes the sigmoid function. Therefore, $g_t, g_i \in (0,1)$.

The refined temporal and image embeddings are obtained through gated residual updates:

$$\begin{aligned} \tilde{\mathbf{f}}_t &= \text{LN}(\mathbf{f}_t + g_t \mathbf{c}_{t \leftarrow i}), \\ \tilde{\mathbf{f}}_i &= \text{LN}(\mathbf{f}_i + g_i \mathbf{c}_{i \leftarrow t}), \end{aligned} \quad (15)$$

where $\text{LN}(\cdot)$ denotes layer normalization. The residual connections preserve the original modality-specific features, while the scalar gates regulate the injection strength of the cross-modal context, thereby reducing the risk that unconstrained feature transfer excessively interferes with the original representations.

Since the two refined embeddings lie in the same d -dimensional shared space and are both layer-normalized, mean aggregation is used to obtain the refined multimodal representation:

$$\mathbf{f}_m = \frac{1}{2}(\tilde{\mathbf{f}}_t + \tilde{\mathbf{f}}_i) \in \mathbb{R}^d. \quad (16)$$

The final fused representation supplied to the evidential decision head is

$$\mathbf{z} = [\mathbf{f}_t; \mathbf{f}_i; \mathbf{f}_m] \in \mathbb{R}^{3d}. \quad (17)$$

This representation simultaneously preserves the original temporal features, the original image features, and the bidirectionally refined multimodal features. The original embeddings provide the decision module with modality-specific information, while the refined representation captures the complementary relationships and consistency between the two modalities.

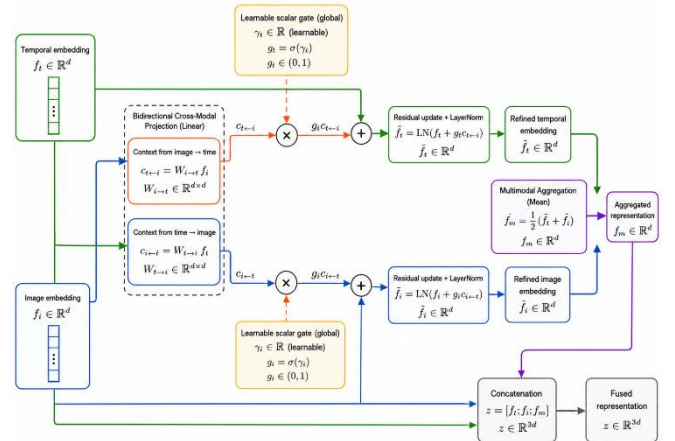


Fig. 2. Gated bidirectional cross-modal refinement module. The temporal and image embeddings are first projected into two directional cross-modal context vectors. Learnable scalar gates regulate the strengths of the corresponding residual updates, after which layer normalization and mean aggregation produce the refined multimodal representation.

E. Dirichlet Evidential Decision Modeling

A conventional Softmax classifier always normalizes the network outputs into class probabilities and may therefore produce highly concentrated predictions even when the input provides only limited discriminative information. To jointly

obtain the spectrum-occupancy probability and an evidence-based uncertainty estimate, a Dirichlet evidential decision head is introduced after the fused representation [12].

Given the fused representation $\mathbf{z}$ defined in (17), a decision MLP first produces a high-level decision feature:

$$\mathbf{h} = \text{MLP}_c(\mathbf{z}) \in \mathbb{R}^{256}. \quad (18)$$

The evidential output layer then generates two outputs corresponding to the spectrum-idle state H_0 and the spectrum-occupied state H_1 :

$$\mathbf{o} = \mathbf{W}_e \mathbf{h} + \mathbf{b}_e \in \mathbb{R}^K, K = 2. \quad (19)$$

The outputs are transformed into nonnegative class evidence using the Softplus function:

$$e_k = \text{softplus}(o_k) + \varepsilon, k \in \{0, 1\}. \quad (20)$$

where e_0 and e_1 denote the evidence supporting the spectrum-idle and spectrum-occupied states, respectively. The corresponding Dirichlet concentration parameters and total strength are defined as

$$\alpha_k = e_k + 1, S = \sum_{k=0}^{K-1} \alpha_k, K = 2. \quad (21)$$

The expected class probability and evidence-based uncertainty are then computed as

$$p_k = \frac{\alpha_k}{S}, u = \frac{K}{S}. \quad (22)$$

When the total evidence is limited, S remains small and u increases accordingly. In contrast, when the multimodal representation provides strong support for a particular spectrum state, S increases and u decreases.

For binary spectrum sensing, the spectrum-occupancy probability is given by

$$p(H_1 | \mathbf{x}) = p_1 = \frac{\alpha_1}{\alpha_0 + \alpha_1}. \quad (23)$$

In this work, p_1 is used for subsequent threshold-based detection, while u serves as an indicator of evidence sufficiency for evaluating prediction reliability. The evidential output layer, decision MLP, bidirectional cross-modal refinement module, and dual-branch feature encoders are jointly optimized under a unified objective.

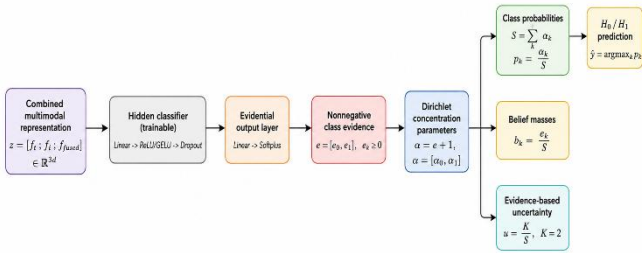


Fig. 3. Dirichlet evidential decision head. The fused representation is successively processed by the decision MLP and evidential output layer to generate nonnegative class evidence for H_0 and H_1 . The class evidence is then converted into Dirichlet concentration parameters, from which the spectrum-occupancy probability and evidence-based uncertainty are derived.

F. End-to-End Training Objective and Decision Rule

To jointly optimize multimodal feature extraction, cross-modal refinement, and evidential decision making, a composite objective combining evidential classification, class-aware focal reweighting, non-target evidence regularization, and pairwise ranking is adopted to balance classification performance, evidence reliability, and positive-negative score separability.

For the i -th sample in a mini-batch, the basic evidential loss is defined as

$$\mathcal{L}_{\text{EDL}}^{(i)} = \sum_{k=0}^{K-1} \left[(y_{i,k} - p_{i,k})^2 + \frac{p_{i,k}(1 - p_{i,k})}{S_i + 1} \right]. \quad (24)$$

where the squared-error term constrains the class prediction, while the variance term accounts for the uncertainty of the Dirichlet prediction. To emphasize difficult samples and spectrum-occupied observations, a class-aware focal factor is applied:

$$\mathcal{L}_{\text{focal}} = \frac{1}{B} \sum_{i=1}^B w_{y_i} (1 - p_{i,y_i})^\gamma \mathcal{L}_{\text{EDL}}^{(i)}. \quad (25)$$

where B is the mini-batch size. The class weights and focusing parameter are set to $w_0 = 1.0$, $w_1 = 1.15$, and $\gamma = 1.5$, respectively.

To suppress unsupported evidence assigned to non-target classes, the adjusted concentration vector is constructed as

$$\tilde{\alpha} = \mathbf{y} + (1 - \mathbf{y}) \odot \alpha, \quad (26)$$

and regularized toward a uniform Dirichlet prior:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}[\text{Dir}(\tilde{\alpha}) \parallel \text{Dir}(\mathbf{1})]. \quad (27)$$

Equation (26) sets the concentration parameter of the ground-truth class to one, allowing the KL term to primarily regularize evidence assigned to the non-target class.

To improve score separability between spectrum-occupied and spectrum-idle observations, the sensing score is defined as

$$s(\mathbf{x}) = \log p_1(\mathbf{x}) - \log p_0(\mathbf{x}). \quad (28)$$

When both classes are present in a mini-batch, the pairwise ranking loss is computed as

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}| \|\mathcal{N}\|} \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{N}} \max[0, m - (s_i - s_j)], \quad (29)$$

where $\mathcal{P}$ and $\mathcal{N}$ denote the spectrum-occupied and spectrum-idle sample sets, respectively, and the ranking margin is set to $m = 0.20$.

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (30)$$

where $\lambda_{\text{KL}} = 0.003$ and $\lambda_{\text{rank}} = 0.12$. This objective jointly optimizes the dual-branch encoders, gated bidirectional cross-modal refinement module, decision MLP, and evidential output layer.

During inference, the final sensing decision is obtained by comparing the spectrum-occupancy probability with a threshold τ :

$$\hat{y} = \begin{cases} 1, & p(H_1 | \mathbf{x}) \geq \tau, \\ 0, & p(H_1 | \mathbf{x}) < \tau, \end{cases} \quad (31)$$

Candidate thresholds are searched on the validation set from 0.05 to 0.95 with an interval of 0.005. The threshold score is defined as

$$J(\tau) = 0.20 \text{ ACC} + 0.28 \text{ F1} + 0.12 \text{ Precision} + 0.28 P_d - 0.06 P_{\text{fa}} - 1.2 \max(0, P_{\text{fa}} - 0.035). \quad (32)$$

The threshold that maximizes $J(\tau)$ is selected for final detection. An additional soft penalty is imposed when P_{fa} exceeds 3.5%. The threshold is determined exclusively on the validation set and remains fixed for all test sets.

IV. PERFORMANCE EVALUATION

This section evaluates the proposed framework from three perspectives: in-distribution sensing performance, generalization to unseen SNR conditions, and blind sensing of unseen modulation formats. The low-SNR transition behavior is further examined to determine whether improved detection is achieved without excessive false alarms.

A. Experimental Setup

(1) Spectrum-Sensing Dataset Construction

RadioML 2018.01A contains 2,555,904 complex baseband sequences from 24 modulation formats, each comprising 1,024 IQ samples with SNRs from -20 to 30 dB in 2-dB increments. To construct a binary spectrum-sensing benchmark, modulated observations are treated as positive samples, whereas synthetic complex AWGN and RadioML samples at -20 dB are used as negative samples. The latter serve as noise-dominated hard negatives and represent an experimental labeling convention rather than a strict physical realization of H_0 .

Stratified sampling is performed over modulation format and SNR, with 76 observations selected from each combination, resulting in 47,424 RadioML samples. Fourteen modulation formats are used for training and in-distribution evaluation, while the remaining ten are reserved for unseen-modulation testing. The evaluation includes in-distribution validation, unseen-SNR testing, and unseen-modulation testing, with the detailed data partition summarized in Table 1.

Table 1. Dataset partitions and evaluation protocols.

Partition	Positive samples	Negative samples	Primary objective
Training	9,363 known-format samples, $[-10, 10]$ dB	851 hard negatives and 8,512 AWGN samples	Model optimization
Validation	2,341 known-format samples, $[-10, 10]$ dB	213 hard negatives and 2,128 AWGN samples	Model selection and threshold calibration
Test Set 1	14,896 known-format samples, $[-18, -12] \cup [12, 30]$ dB	14,896 AWGN samples	Unseen-SNR generalization
Test Set 2	19,000 unseen-format samples, $[-18, 30]$ dB	760 hard negatives and 18,240 AWGN samples	Blind-modulation generalization

Known formats: BPSK, QPSK, 8PSK, 16PSK, OOK, 4ASK, 8ASK, 16QAM, 32QAM, 64QAM, AM-DSB-SC, AM-SSB-SC, GMSK, and OQPSK.

Unseen formats: 32PSK, 16APSK, 32APSK, 64APSK, 128APSK, 128QAM, 256QAM, AM-SSB-WC, AM-DSB-WC, and FM. All SNR intervals use a step size of 2dB.

Test Set 1 retains the modulation families used during training but evaluates only SNR values outside the training interval. It therefore isolates SNR-domain generalization. Test Set 2 contains positive samples exclusively from modulation formats excluded from training and evaluates whether the detector learns general signal-presence characteristics rather than modulation-specific templates.

(2) Baseline Methods

All baseline models are trained and evaluated using the same data partitions and evaluation protocol as the proposed method.

- **CNN-based method** ^[21]: An image-only ResNet18 processes the same eight signal-derived image modalities as the proposed method to evaluate the spectral, angular-correlation, transition, and recurrence information without temporal modeling or cross-modal interaction.

- **BiLSTM-based method** ^[8]: A temporal-only two-layer BiLSTM models the bidirectional dependencies of four standardized sequential features to evaluate the effectiveness of temporal waveform evolution.

- **BCAM-Net** ^[9]: The reproduced BCAM-Net combines a BiGRU-based temporal branch, a CWT-based time-frequency branch, and bidirectional multihead cross-modal interaction, serving as the principal multimodal baseline for evaluating the proposed gated bidirectional cross-modal refinement method.

(3) Implementation Details

During training, phase rotation, circular time shifting, carrier-frequency offset, IQ imbalance, amplitude scaling, and additive-noise perturbation are applied. Additional low-SNR positive samples are generated to improve weak-signal detection, whereas no data augmentation is used for validation or testing.

The model is trained using AdamW with a batch size of 64 for a maximum of 55 epochs. A learning-rate warm-up is applied during the first five epochs, followed by cosine decay. Exponential moving averaging, gradient clipping, and early stopping with a patience of 12 are also employed. The EMA checkpoint obtained at epoch 37 is used for evaluation. The complete model contains approximately 11.89 million trainable parameters.

Training is performed using the joint objective defined in (30). The decision threshold of each model is determined exclusively on its corresponding validation set and remains fixed during testing. The resulting threshold for the proposed model is 0.560. All experiments are implemented in PyTorch and conducted on an NVIDIA Tesla T4 GPU.

(4) Evaluation Metrics

The reported metrics include accuracy, F1-score, precision, detection probability P_d , false-alarm probability P_{fa} , and AUC. The sensing-specific metrics are

$$P_d = \frac{TP}{TP + FN}, P_{fa} = \frac{FP}{FP + TN} \quad (33)$$

AUC evaluates threshold-independent ranking quality, whereas P_d and P_{fa} characterize the detector at the selected operating point. In practical spectrum sensing, a high P_d is useful only when P_{fa} remains sufficiently low.

For the proposed evidential model, the uncertainty mass is calculated as

$$u = \frac{K}{\sum_{k=0}^{K-1} \alpha_k}, \quad (34)$$

where $K = 2$ is the number of sensing classes. Mean uncertainty and the uncertainties of correct and incorrect predictions are additionally reported to assess whether the evidential output can identify unreliable decisions.

B. Comparison Experiments

Table 2 compares the proposed method with BCAM-Net, CNN, and BiLSTM on the validation set, the unseen-SNR test set, and the unseen-modulation test set.

(1) Overall Quantitative Comparison

The proposed method achieves the highest ACC, F1-score, and AUC on all three evaluation sets. On the validation set, it attains an ACC of 0.9507, an F1-score of 0.9490, and an AUC of 0.9805, together with $P_d = 0.9188$ and $P_{fa} = 0.0175$. Compared with BCAM-Net, the proposed method improves ACC, F1-score, and AUC by 8.65, 10.09, and 5.49 percentage points, respectively. These results indicate that gated bidirectional cross-modal refinement and evidential training enable more effective integration of heterogeneous temporal and image features.

Although CNN achieves a relatively high P_d , its P_{fa} reaches 0.2050, indicating evident over-detection. In contrast, the proposed method improves occupied-spectrum detection while maintaining a low false-alarm rate, thereby providing a more balanced operating point.

(2) Generalization Under Unseen SNRs

On Test Set 1, the proposed method achieves an ACC of 0.8530, an F1-score of 0.8316, and an AUC of 0.8752, while maintaining $P_d = 0.7260$ and $P_{fa} = 0.0200$.

Table 2. Performance comparison of different models on three evaluation sets.

Set	ACC				F1-score				Precision			
	Ours	BCAM	CNN	BiLSTM	Ours	BCAM	CNN	BiLSTM	Ours	BCAM	CNN	BiLSTM
Validation	0.9507	0.8642	0.8379	<u>0.9231</u>	0.9490	0.8481	0.8446	<u>0.9180</u>	<u>0.9813</u>	0.9616	0.8112	0.9834
Test 1	0.8530	0.8206	0.7862	<u>0.8526</u>	0.8316	0.7896	0.7850	<u>0.8304</u>	<u>0.9732</u>	0.9545	0.7895	0.9778
Test 2	0.8890	0.8362	0.8075	<u>0.8808</u>	0.8779	0.8114	0.8106	<u>0.8674</u>	<u>0.9752</u>	0.9559	0.7978	0.9776
Set	Recall (Pd)				Pfa				AUC			
	Ours	BCAM	CNN	BiLSTM	Ours	BCAM	CNN	BiLSTM	Ours	BCAM	CNN	BiLSTM
Validation	0.9188	0.7587	<u>0.8808</u>	0.8607	<u>0.0175</u>	0.0303	0.2050	0.0145	0.9805	0.9256	0.9304	<u>0.9644</u>
Test 1	<u>0.7260</u>	<u>0.6733</u>	0.7805	0.7216	<u>0.0200</u>	0.0321	0.2081	0.0164	0.8752	0.8590	0.8615	<u>0.8650</u>
Test 2	<u>0.7983</u>	0.7049	0.8238	0.7795	<u>0.0203</u>	0.0325	0.2088	0.0178	0.9166	0.8922	0.8923	<u>0.9074</u>

The best and second-best results are shown in bold and underlined, respectively.

Compared with BCAM-Net, the ACC, F1-score, and AUC are improved by 3.24, 4.20, and 1.62 percentage points, respectively. This demonstrates that the proposed method preserves stronger positive-negative score separability under SNR conditions outside the training range.

CNN obtains a higher P_d of 0.7805, but its P_{fa} increases to 0.2081. By contrast, the proposed method keeps the false-alarm rate at approximately 2%, indicating that its robustness does not rely on an aggressively reduced decision threshold, but instead benefits from more reliable weak-signal representation and cross-modal feature modeling.

(3) Low-SNR Transition and Blind-Modulation Generalization

All positive samples in Test Set 2 belong to modulation types excluded from training, making this set suitable for evaluating generalization under blind modulation conditions. The proposed method achieves an ACC of 0.8890, an F1-score of 0.8779, an AUC of 0.9166, $P_d = 0.7983$, and $P_{fa} = 0.0203$. Compared with BCAM-Net, the ACC, F1-score, P_d , and AUC increase by 5.28, 6.65, 9.34, and 2.44 percentage points, respectively.

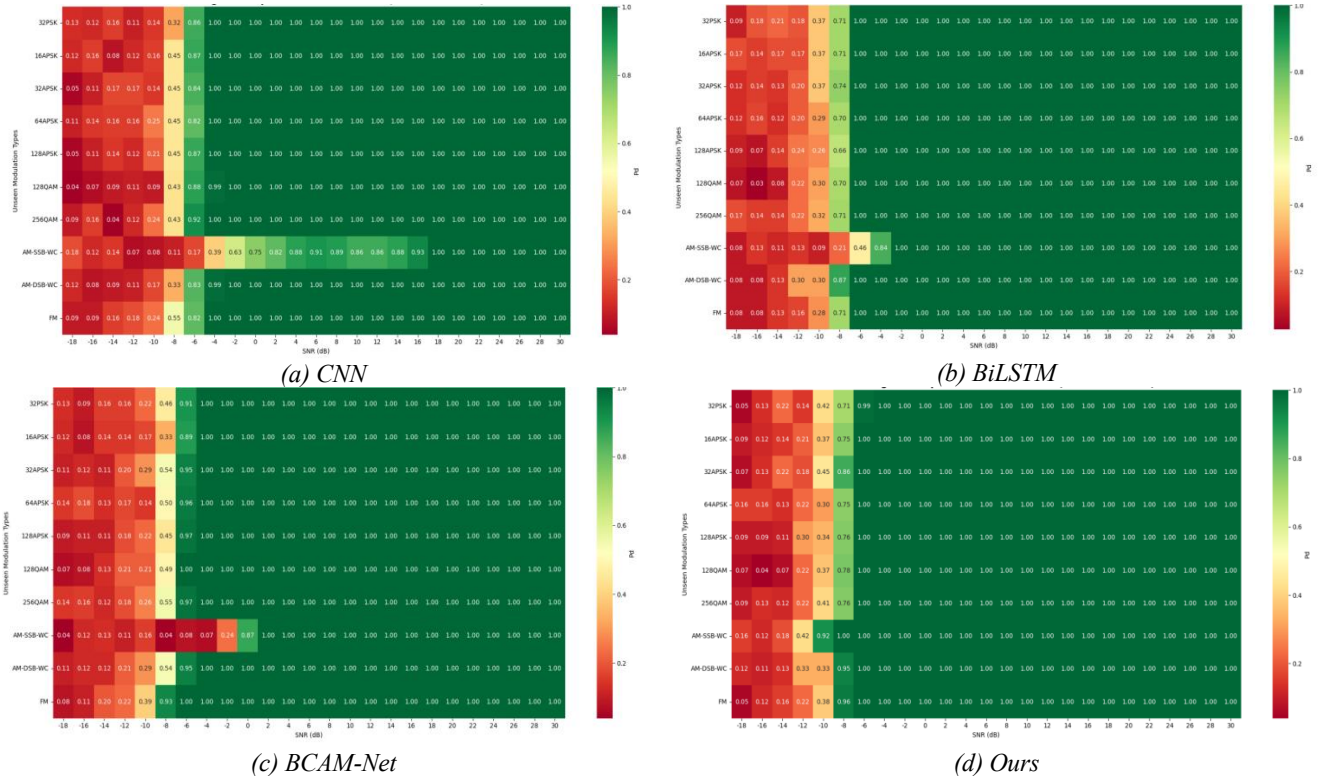


Fig. 4. Modulation-wise detection probability P_d heatmaps on Test Set 2 for unseen modulation formats: (a) CNN, (b) BiLSTM, (c) BCAM-Net, and (d) the proposed method. All panels use identical modulation and SNR orders, a shared color scale, and thresholds calibrated on the corresponding validation sets.

As shown in Fig. 4, the proposed method exhibits more consistent detection performance across unseen modulation types and SNR levels. Although CNN achieves a slightly higher P_d , its P_{fa} reaches 0.2088. In comparison, the proposed

method reduces the false-alarm rate by 18.85 percentage points while maintaining a competitive detection probability. This result suggests that the proposed framework learns more

transferable signal-presence characteristics rather than relying primarily on modulation patterns observed during training.

Overall, the main advantage of the proposed method is not merely a higher recall, but a consistently better balance among P_d , P_{fa} , and AUC under in-distribution, unseen-SNR, and unseen-modulation conditions.

V. CONCLUSION

This paper presented an evidential multimodal framework for reliable binary spectrum sensing under SNR and modulation distribution shifts. Complementary temporal and signal-derived image representations are jointly modeled through dedicated feature encoders and a gated bidirectional cross-modal refinement module, while Dirichlet evidential learning provides spectrum-occupancy predictions together with an evidence-based uncertainty estimate.

Experiments under in-distribution validation, unseen-SNR testing, and unseen-modulation testing demonstrate that the proposed framework consistently achieves the best ACC, F1-score, and AUC among the compared methods while maintaining a low false-alarm probability. The results further show that reliable spectrum sensing should not be assessed solely by in-distribution classification accuracy. Robust sensing requires stable score separability, controlled false alarms, and transferable signal-presence detection under distribution shifts. From this perspective, the proposed framework provides a promising approach to trustworthy physical-layer sensing by combining complementary signal representations with uncertainty-aware decision modeling.

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